

Fill in the table below with the correct answer

(50 Pts)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
Q23	Q24	Q25								

1- If $y'' = \delta(t+1)$, $y(0) = 0$, $y'(0) = -1$, such that $\ell\{y(t)\} = Y(s)$, then $Y(1) =$

- a) $e^{-1} - 1$ b) $e - 1$ c) $e + 1$ d) $e^{-1} + 1$

2- The particular solution of $yy' = x+1$, $y(0) = 1$ is $y^2 =$

- a) $x^2 + 2x - 1$ b) $x^2 + 2x + 2$ c) $x^2 + 2x$ d) $x^2 + 2x + 1$

3- If $(y^3 + kxy^4 - 2x)dx + (3xy^2 + 20x^2y^3)dy = 0$, then the value of k that makes the D.E exact is

- a) -10 b) 4 c) 40 d) 20

4- If $X_1 = \begin{pmatrix} e^{-t} \\ e^t \end{pmatrix}$, $X_2 = \begin{pmatrix} 3e^{-5t} \\ -e^{-5t} \end{pmatrix}$ are two solutions of the homogeneous part of

$X' = AX + \begin{pmatrix} 6 \\ 1 \end{pmatrix}$ such that the particular solution $X_p = X_1u_1 + X_2u_2$, then $u_1 =$

- a) $\frac{9}{4}e^t$ b) $\frac{9}{4}e^{-t}$ c) $\frac{1}{4}e^{5t}$ d) $\frac{25}{4}e^{5t}$

5- The suitable form for y_p of $y'' + y' - 4y = 2e^{-x} + \cos x$ is

- a) $Ae^{-x} + B\cos x + C\sin x$ b) $Axe^{-x} + B\cos x + C\sin x$
 c) $Axe^{-x} + B\cos x$ d) $x e^{-x} (A + Bx) + C\sin x + D\cos x$

6- The Laplace transform of $\{t^2 \delta(t-3)\}$ equals

- a) $9e^{3s}$ b) $9e^{-3s}$ c) $-3e^{-3s}$ d) $3e^{3s}$

7- If the variation of parameters method is used to find $y_p = y_1 u_1 + y_2 u_2$ for $x^2 y'' - 3xy' + 3y = 2x^4 e^x$, such that $y_1 = x, y_2 = x^3$ then $u_2 =$

- a) $-x^2 e^x + 2x e^x - 2e^x$ b) $-e^{-x}$ c) $-2e^x$ d) e^x

8- The singular solution of $y = xp + (4-p)^2$, where $p = y'$, is $y =$

- a) $-4x + \frac{3x^2}{4}$ b) $4x - \frac{x^2}{4}$ c) $-4x + \frac{x^2}{4}$ d) none

9- The inverse Laplace of $\left\{ \frac{s^2 - 1}{(s+1)^5} \right\}$ equals

- a) $\frac{1}{2} t^3 e^{-t} - \frac{1}{3} t^3 e^{-t}$ b) $t^2 e^{-t} - \frac{1}{2} t^3 e^{-t}$ c) $\frac{1}{2} t^3 e^{-t} - \frac{1}{6} t^4 e^{-t}$ d) none

10- Assume that $F(s) = t \left\{ \sinh(\ln t) + \frac{1}{2t} \right\}$, then $F(0.5) =$

- a) -0.5 b) 4 c) -2 d) 2

11- The orthogonal trajectories of the family of curves $e^x + e^{-x} = c$ are

- a) $e^y + e^{-y} = c$ b) $e^y + e^x = c$ c) $e^y - e^{-x} = c$ d) $e^{-y} + e^x = c$

12- The general solution of $y'' - 6y' + 12y = 0$ is

- a) $c_1 e^x + c_2 e^{3x} + c_3 e^{4x}$ b) $c_1 e^{-x} + c_2 e^{3x} + c_3 e^{4x}$
 c) $c_1 e^x + c_2 e^{-3x} + c_3 e^{-4x}$ d) $c_1 e^{-x} + c_2 e^{3x} + c_3 e^{-4x}$

13- If the G.S. of $x^3 y''' + ax^2 y'' + bxy' + cy = 0$ is $y = c_1 x + c_2 x \ln x + c_3 x^2$ then $a =$

- a) 1 b) 0 c) -1 d) 2.

14- The general solution of $(x \tan(\frac{y}{x}) + y)dx - xdy = 0$ is

- a) $x \cos(\frac{y}{x}) = c$ b) $x \sin(\frac{y}{x}) = c$ c) $x \csc(\frac{y}{x}) = c$ d) $x \sec(\frac{y}{x}) = c$.

15- By using $y = \sum_{n=0}^{\infty} a_n x^n$ as the general solution of $y'' + 4y = 0$, then $a_2 =$

- a) $-2a_0$ b) $-\frac{2}{3}a_0$ c) $-4a_0$ d) $-\frac{1}{3}a_0$.

16- If $Y(s) = \ell\{y(t)\} = \frac{s}{s+1}$, and $\ell\{ty'(t)\} = Y_1(s)$, then $Y_1(1) =$

- a) $-\frac{3}{4}$ b) $-\frac{1}{4}$ c) $-\frac{1}{2}$ d) $-\frac{5}{4}$

17- If $f(t) = \ell^{-1}\left\{\frac{e^{2s}}{s^2} - \frac{1}{s^3}\right\}$, then $f(2) =$

- a) -2 b) 2.5 c) 2 d) 0

18- The integrating factor of $xy' + 2xy + 2y = e^{-x} \sin(2x)$ is

- a) $2 \ln(x) + x^2$ b) xe^x c) xe^{2x} d) $x^2 e^{2x}$

19- If $F(s) = \ell\{e^{-3t} f(t)\} = e^{-2(s+3)}$, then $f(t) =$

- a) $u(t+2)$ b) $\delta(t-2)$ c) $\delta(t+2)$ d) $u(t-2)$